

Enhancing Bimodal Communication with AI-Powered Glasses for English and ASL

Improving Real-Time Communication Between Deaf and Hearing Individuals

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ABSTRACT

Effective two-way communication between Deaf and Hard-of-Hearing (D/HoH) individuals and non-signers remains an ongoing challenge, particularly in dynamic, real-world scenarios. In this study, we evaluate a smart glasses-based communication system that leverages speech recognition and Large Language Models (LLMs) to support real-time ASL-English interactions. Our focus is on assessing how hearing individuals perceive and interact with this technology when conversing with a simulated Deaf user. We began with a pilot session involving a Deaf professor whose feedback led to key interface and gesture design improvements. We then conducted a scenario-based study in which hearing participants role-played as restaurant servers and interacted with a researcher using the glasses, followed by a texting-based baseline and comparative surveys. A second study involved structured interviews with 12 participants who watched demonstration videos and reflected on their expectations and experience. Our findings provide insights into usability, timing, and social dynamics, and identify key design opportunities for improving comfort, clarity, and interaction flow in future AR-based communication systems.

INTRODUCTION

Real-time communication between Deaf and Hard-of-Hearing (D/HoH) individuals and non-signers remains a persistent challenge, especially in public or service-oriented settings like restaurants, clinics, and airports. While common strategies such as text messaging, note-passing, or interpreters are often used, these methods can feel impersonal, inconvenient, or socially limiting. Emerging technologies in augmented reality (AR) and Large Language Models (LLMs) offer new possibilities for bridging this gap by enabling more natural, two-way interactions through smart glasses. These glasses display transcribed speech from hearing individuals and generate selectable response options for D/HoH users, facilitating smoother conversations.

This project investigates how hearing individuals perceive and interact with a smart glasses-based ASL communication system powered by an LLM. The system captures spoken

input from hearing users and offers pre-generated responses on the glasses display, which the D/HoH user can select using simple gesture-based inputs. While the core hardware and software—speech recognition, GPT-based language generation, and gesture detection—were developed by our mentor, our focus was on evaluating the interaction experience through a series of structured user studies with hearing participants.

We began with a pilot session involving a Deaf professor, whose feedback on interface readability, gesture recognition accuracy, and cultural sensitivity informed key system adjustments. These included changing font colors for better contrast, efficiency improvements, refining left-handed gesture detection, and replacing certain confirmation cues with more culturally appropriate signals. Next, in Study 1, hearing participants engaged in live roleplay as restaurant servers communicating with a Deaf customer using the glasses. They placed an order, asked follow-up questions, and observed how the system generated and displayed response options. Because real-time sign recognition was not yet implemented, a researcher manually input signed responses during the simulation.

Participants then repeated the same scenario using a texting method as a baseline and completed Likert-scale surveys comparing both experiences across dimensions such as comfort, clarity, engagement, and patience. In Study 2, a new group of participants watched demonstration videos of the AR glasses in use—both third-person and first-person perspectives—and participated in structured interviews exploring ideal system behavior, visual cues, personalization, and perceived social naturalness.

Our contribution lies in understanding how hearing individuals experience, interpret, and respond to AR-based communication systems. By gathering both quantitative and qualitative data on usability, expectations, and social dynamics, we aim to guide future iterations of the technology to better serve real-world D/HoH-hearing interactions in both structured and unpredictable environments.

RELATED WORK

Research on technological sign language interpretation has used sensors, cameras, and hybrid systems, but many still struggle with real-time communication between D/HoH individuals and non-signers. Our Two-Way ASL Communication System aims to solve these issues using smart glasses and AI models for better accessibility.

Wearable Sensor-Based Approaches

Many systems use sensors to track hand movements. Ahmed et al. [1] developed a glove with 17 sensors, while Yetkin et al. [2] designed a lighter ring-based system. These work well but require users to wear special hardware, which isn't always practical.

Vision-Based and Hybrid System

DeepASL [3] uses infrared and deep learning for ASL recognition but struggles with lighting and backgrounds. Kudrinko et al. [4] reviewed various sensor-based methods, noting issues like high costs and slow processing. Some systems combine depth cameras with wearables, but real-time conversation is still a challenge.

Augmentative and Alternative Communication (AAC) System

Valencia et al. [5] explored assistive robots for non-verbal communication, but these focus on enhancing interactions rather than full ASL interpretation.

Context-Aware Computing Systems

Dey [6] defines context as "information that characterizes an entity's situation" and describes context-aware systems as those providing relevant services based on this information. The Context Toolkit architecture he presents supports building applications that respond appropriately to changing environments and user needs. Schilit et al. [7] identify key context elements ("where you are, who you are with, what resources are nearby") and categorize context-aware applications into four types: proximate selection, automatic contextual reconfiguration, contextual information and commands, and context-triggered actions. While these frameworks provide powerful approaches for adaptive computing, they've rarely been applied to sign language translation systems, which typically treat sign recognition as isolated pattern matching rather than considering the full conversational and environmental context.

Dual Speaker-Based Acoustic Gesture Recognition

The DuG paper [8] introduces a Dual Speaker Gesture recognition system that uses two speakers and a microphone to detect hand movements through the Doppler effect, achieving 99.36% accuracy.

Speech-to-Text

The FastSpeech and TTS paper [9, 10] introduce important approaches for text-to-speech synthesis. FastSpeech [9] addresses key challenges through a non-autoregressive approach, achieving 270x speedup, improved robustness via a length regulator and duration predictor, and better controllability over voice speed. The system introduces a novel feed-forward network based on the Transformer architecture that enables parallel generation of mel-spectrograms and implements a duration predictor trained on phoneme durations for explicit control over speech timing. Similarly, [10] focuses on accessibility through Object-Oriented Analysis and Development Methodology (OOADM), using an Expert System and Speech Synthesis Module to help individuals with physical impairments through phonetic component extraction. The system incorporates SAPI and JSAPI support through a Free TTS speech engine, enabling effective control over speech signal generation.

How Our Work is Different

1. **Real-Time Two-Way Interpretation:** Most systems only convert ASL to text/speech, but ours interprets both ways for smooth conversation.
2. **Smart Glasses Integration:** AR glasses provide a more practical, comfortable solution, unlike gloves or external cameras.
3. **AI-Powered Context Awareness:** Our system uses AI to improve interpretation accuracy and natural language flow.
4. **Gesture-Based Controls:** Users can refine interpretations with simple hand gestures, reducing miscommunication.
5. **Our system implements Dey's context-aware framework for bidirectional ASL-English communication, considering conversation history, location, participants, and topic to generate more accurate and contextually appropriate interpretations.**
6. **While DuG demonstrates the effectiveness of dual-source gesture recognition, our approach differs by using computer vision and specialized sensors rather than acoustic signals.**
7. **While both works inform our approach, our system will integrate speech input using microphones alongside computer vision and LLMs, creating a multi-modal interpretation system that improves upon the single-modality approaches of [9] and [10].**

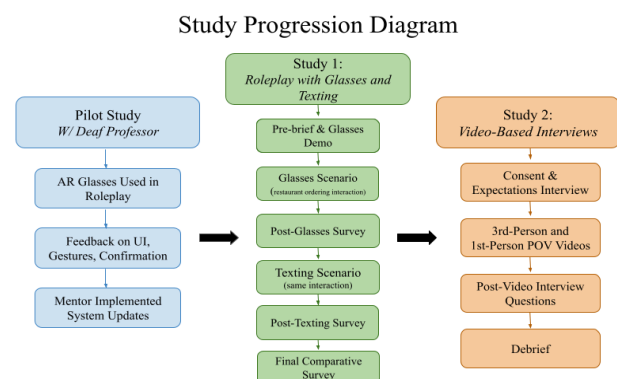
By overcoming these challenges, our system offers a more user-friendly and efficient solution for real-time ASL communication.

DESIGN AND IMPLEMENTATION

This project evaluates an augmented reality (AR) smart glasses-based system that supports two-way communication between Deaf and Hard-of-Hearing (D/HoH) individuals and hearing non-signers. While the underlying hardware and software were developed by a separate research team, our work centers on evaluating the usability, communication flow, and social reception of the system from the perspective of hearing users. We designed and conducted a series of studies, including a pilot test, a scenario-based roleplay experiment (Study 1), and a follow-up video-based interview study (Study 2), to gather feedback and inform iterative design improvements.

The studies were intentionally sequenced to reflect a natural user-testing progression. We began with a pilot study involving a Deaf professor who offered critical feedback on accessibility, gesture clarity, and interface usability. Based on her input, our mentor—the developer of the glasses—made early system updates, including improved font visibility, refined gesture recognition, and culturally sensitive confirmation cues. Study 1 followed, where hearing participants engaged in live roleplay using the glasses and then a baseline texting condition, providing comparative survey and observational data. Finally, in Study 2, a new group viewed videos of the glasses in use and participated in structured interviews exploring their expectations, interpretations, and design suggestions. These findings are now being shared with the mentor to inform further system improvements. Figure 1:

Figure 1. Study progression diagram showing the sequence and structure of the pilot study, Study 1 (live roleplay), and Study 2 (video-based interviews), along with their respective components and data collection stages.



Pilot Study: Deaf Professor Feedback [MIDTERM]

To ensure that the system met accessibility needs and cultural expectations, we conducted an early pilot test with a Deaf professor at Cornell University. She wore the smart glasses and completed a restaurant-style ordering scenario with a hearing individual, supported by an interpreter. Her feedback highlighted several areas for improvement:

1. **Visual Accessibility:** She found the original green font difficult to read and suggested yellow text for better contrast.
2. **Gesture Clarity:** Detection failed when she used her left hand while making selections with her right. The system was not adequately tuned for left-handed gestures.
3. **Cultural Appropriateness:** The thumbs-up gesture used as a confirmation felt inappropriate in a Deaf cultural context; she recommended using a more neutral or context-aware confirmation.
4. **Efficiency:** Originally, the generated options were generated solely from signed input. She suggested only requiring a signed response if an initial option set does not match the intended message.
5. **System Transparency:** She emphasized the importance of the glasses indicating whether the system is actively listening, processing, or speaking to reduce user confusion.

Following this pilot, our mentor made a series of design updates to the system, modifying font color, gesture detection thresholds, and initial logic for response generation.

Unfortunately, due to timing and budget constraints, we could not secure a certified interpreter to facilitate the conduction of this study. Adriana, an ASL student, informally interpreted some dialogue. The team also utilized text methods during this portion.

Study 1: Scenario-Based Roleplay with Smart Glasses and Texting [MIDTERM]

Study 1 focused on how hearing individuals perceived and interacted with a D/HoH individual using the smart glasses in a real-time restaurant setting. Four participants engaged in a live dialogue with a researcher role-playing as a Deaf customer using the AR glasses.

Study Protocol and Materials

Participants were first welcomed, informed about the session's purpose, and gave written consent. They then

received a live demo of how the glasses worked from one of our team members. Once they demonstrated understanding of the interface and gesture selection system, they were introduced to a second team member, who knows ASL but pretended to be Deaf for the purpose of simulating a realistic interaction. The participant was unaware that this team member could hear.

In the primary interaction, the participant acted as a restaurant server. The Deaf character placed a complex food order:

- 1. “I’d like the ribeye steak, medium rare, with mushroom sauce.
- 2. “Fries with no salt, please.”
- 3. “Coke with extra ice.”

Follow-up questions simulated a typical dining experience:

- 1. “How was your meal?” (Response: “It was overcooked.”)
- 2. “Would you like a to-go box?”
- 3. “Do you want to pay with card or cash?”

The smart glasses used a GPT-based LLM to generate 3–4 suggested responses based on the server’s spoken input, which were displayed on the glasses screen. The user selected a response using 1–4 finger gestures. Raising five fingers triggered a reset, allowing a signed message to be manually entered by a researcher (as sign recognition was not yet functional).

After the glasses interaction, the participant completed a Post-Glasses Survey using a 1–7 Likert scale to assess clarity, comfort, social engagement, and patience. Then, the same roleplay was repeated using a texting method, where the participant passed a phone back and forth with the D/HoH character. A Post-Texting Survey followed, using the same questions. Lastly, a Final Survey asked for comparative preferences, qualitative impressions, and suggestions.

Likert Scale Meaning (1–7)

Participants rated each condition on a scale where:

- 1. 1 = Negative Extreme (e.g., Very Confusing, Very Impersonal)
- 2. 7 = Positive Extreme (e.g., Very Clear, Very Engaging)

These ratings helped us compare experiences across conditions.

Study 1 Results: Survey Data

Figure 2. Here is the bar chart visualizing Study 1's average Likert scale results, comparing user experiences with Smart Glasses vs. Texting across five communication dimensions: Ease of Communication, Clarity of Message, Comfort, Social Connection, and Patience. The y-axis reflects the average participant score on a 1–7 scale, where: 1 = very negative (e.g., very difficult, very confusing, very impersonal), and 7 = very positive (e.g., very easy, very clear, very engaging)

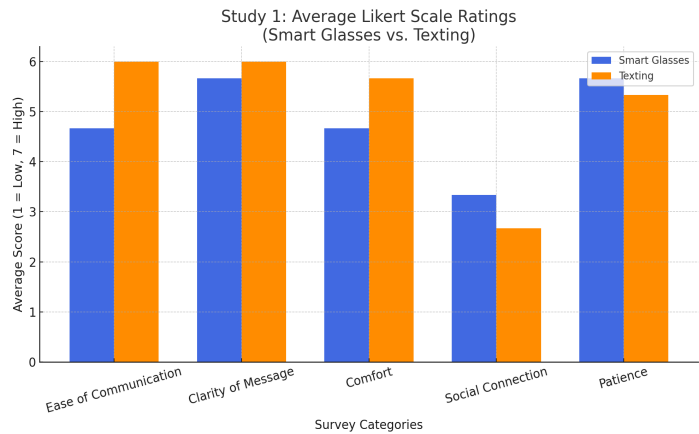


Figure 2 illustrates that while texting was generally rated as easier and more comfortable, the smart glasses were perceived as more engaging and slightly clearer in communication by some participants. These results suggest that although the glasses are still in early development, they hold promise for fostering more natural social interactions compared to traditional methods, especially in real-world scenarios like restaurants.

Table 1. To complement the bar chart visualization, the following table summarizes key differences in participants’ experiences across five communication dimensions: Ease of Communication, Clarity, Comfort, Social Connection, and Patience. For each category, we report the average Likert score for Smart Glasses and Texting, the difference between the two, and a representative participant quote. This comparative format provides both a quantitative overview and a qualitative grounding of user impressions.

Communication Dimension	Smart Glasses Avg.	Texting Avg.	Difference	Sample Quote
Ease of Communication	5.5	6.5	↓ 1.0	"Texting was faster."
Clarity of Message	6.0	6.2	↓ 0.2	"Both were clear, but text is safer."
Social Connection	6.1	4.5	↑ 1.6	"The glasses felt more human."
Comfort	5.0	6.8	↓ 1.8	"I was nervous using the glasses."
Patience Required	6.0	5.5	↑ 0.5	"The delay was fine."

The results of Table 1. highlight where the AR glasses showed early promise, particularly in fostering social connection and authentic engagement, despite being slightly less comfortable or intuitive than texting. For instance, while texting was rated as easier and more familiar, several participants described the glasses as more “conversational” and “human.” Conversely, concerns around system clarity and user control were reflected in lower comfort scores, with one participant noting uncertainty about system responsiveness.

This mixed feedback underscores the need for design improvements in usability and clarity, while affirming the system’s potential to support more natural social interaction, especially in structured scenarios like restaurants or customer service contexts.

In addition to the visual data, open-ended responses revealed mixed preferences:

1. One participant praised the glasses as “more conversational and natural” than texting.
2. Another noted discomfort due to unclear system states and the lack of visible facial expression.
3. One participant preferred texting but said they would favor glasses in social settings if the system improved response timing and customization.

Study 2: Video-Based Interview Study [FINAL]

To gather broader insight into how hearing users interpret and anticipate AR-based communication systems, we conducted a second study with 12 participants. This time, participants observed the glasses in use via recorded videos and shared their expectations and feedback in a semi-structured interview.

Study Protocol and Materials

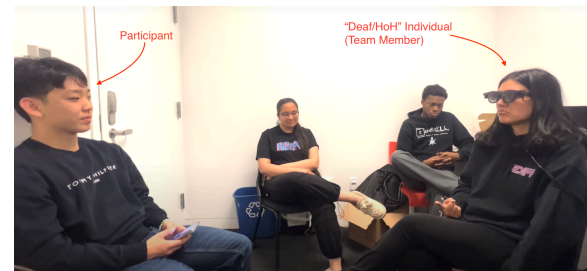
Each session began with informed consent and a general introduction to the study’s goal. Participants were asked pre-video questions, such as:

1. “What do you expect these AR glasses to do?”
2. “Have you ever communicated with a Deaf or Hard-of-Hearing person?”

Next, they watched two videos:

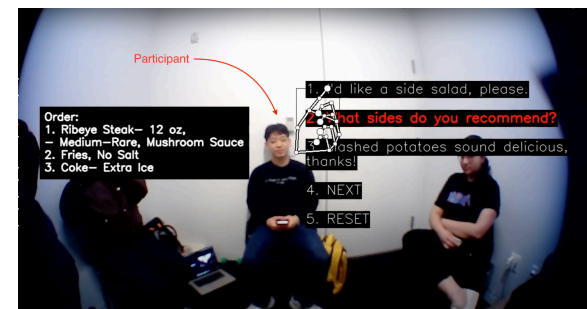
1. Third-person video showing the glasses in action during a restaurant interaction. [\[Video link\]](#)

Figure 3(a). Third-person perspective showing live interaction with the Deaf participant.



2. First-person POV video simulating the user experience of the Deaf individual [\[Video link\]](#)

Figure 3(b). First-person POV showing interface and gesture-based response options.



Afterward, participants answered a structured set of questions exploring:

1. Mental Models: “Did the glasses work the way you expected?”
2. Visual Cues: “What kinds of indicator lights or gestures would help you know what’s happening?”
3. Timing: “Would you prefer faster or slower responses?”
4. Social Dynamics: “Would you feel more connected using glasses vs texting?”

Key Findings

Table 2. The table below summarizes the key design dimensions that emerged from our second study, with

Design Dimension	Common Feedback	Sentiment	Representative Quotes
Visual Indicators	Preferred color-coded signals (Green = understood, Yellow = processing, Red = wait)	Mostly Positive	“Green for understood, red for needs more time.”
Timing & Responsiveness	Preferred ~2-5 second response delay; real-time OK if accurate	Mixed to Positive	“Slower but accurate is better... within five seconds is fine.”
Naturalness of Speech	Concern over robotic tone; preference for natural inflection	Mostly Negative	“Make the voice sound more human... questions should rise in tone.”
Interaction Flow	Smooth in structured scenarios (e.g. restaurants); concerns in chaotic settings	Mixed	“In noisy places, it might not know who to listen to.”
Social Comfort	Increased willingness to engage with DHH individuals	Positive	“I’d feel less nervous and more open to talking with someone Deaf.”
Preference over Texting	More authentic and warm than texting or writing	Positive	“It’s more human than passing notes or texting back and forth.”
System Limitations	Concerns about translation errors or lack of personalization	Mixed to Negative	“What if it misinterprets a gesture near the face?”

participant sentiment and representative quotes for each. These categories encapsulate the major usability and communication considerations surfaced during the video-based interviews:

The following subsections expand upon these themes, synthesizing both qualitative insights and repeated patterns across participants.

Visual Indicators

Visual feedback was a recurring theme across interviews. Most participants supported color-coded signals to communicate the system's internal state, especially for the hearing communication partner who might not be sure when to speak or wait. The most intuitive pattern, cited by six participants, was green for "understood" or "ready to listen," yellow for "processing," and red for "responding" or "please wait." One participant noted that without these signals, "it's hard to tell if they're still reading or already done." While no one explicitly objected to visual cues, several participants stressed that such indicators must remain subtle to avoid drawing too much attention in public spaces. Design suggestions included a small LED near the lens rim or side, with smooth transitions between states.

Timing & Responsiveness

Most participants agreed that a short pause, about 2 to 5 seconds, was acceptable and even desirable, provided it ensured accuracy. Several felt that an immediate response would feel robotic or rushed, while others emphasized that timing preferences might vary by context. One suggestion was to allow users to toggle between "fast" and "accurate" response modes based on the social setting. For example, a restaurant exchange might benefit from speed, while a doctor's office might demand more precision. Though a few participants compared the ideal delay to tools like Siri or Google Assistant, many emphasized that "real-time" wasn't the goal, clarity and flow were.

Naturalness of Speech

A frequent point of criticism was the AI voice used in the demo videos. Participants described it as too robotic, lacking natural inflection, and "not human enough to make a real conversation feel smooth." Several noted that emotional tone and emphasis are crucial, especially for interpreting questions or conveying empathy. Some proposed letting users choose a voice from a library of options (e.g., friendly, neutral, enthusiastic), while others imagined a more advanced future system that could detect the emotional content of ASL input and select matching vocal inflection. These findings suggest that synthetic speech remains a

major area for refinement if conversational authenticity is the goal.

Interaction Flow

Participants praised the structured interaction shown in the restaurant videos, especially the system's use of pre-generated options. The ability to simply sign a number to select a response was seen as both efficient and intuitive. However, many also pointed out that such a design would be limiting in more open-ended conversations. In more spontaneous interactions, like at a museum, airport, or social event, participants worried that too rigid a system might "get in the way." Several recommended a hybrid design where the glasses first offer a few predicted replies but allow the Deaf user to sign a custom response if none fit. One participant suggested adding a "skip" or "custom reply" gesture to override preset options when needed.

Social Comfort

The glasses' design was generally seen as socially empowering. Multiple participants mentioned that they would feel more comfortable approaching or responding to a Deaf or Hard-of-Hearing person using these glasses. "I'd feel less nervous and more open to talking," one participant said, capturing a common sentiment that the technology reduces the intimidation factor of cross-communication. Others noted that it removes the awkwardness of guessing turn-taking cues or feeling unsure when to speak. This perceived confidence boost suggests that beyond functionality, the glasses play an important affective role in lowering the barriers to initiating inclusive interaction.

Preference over Texting

Nearly every participant preferred the glasses to traditional text-based methods like phone typing or paper notes. This preference stemmed from both social and logistical reasons. First, the glasses offered a more conversational rhythm, which participants described as "human" and "flowing." Second, participants noted that texting while face-to-face can feel impersonal or awkward. One participant said, "It's more human than passing notes back and forth." This feedback reinforces the value of creating systems that emulate real-time dialogue rather than asynchronous messaging.

System Limitations

Despite their overall optimism, participants raised valid concerns about the limitations of the current system. Several mentioned the risk of gesture misinterpretation, particularly for signs made near the face or in crowded environments. Others questioned how the system would know who to

listen to in noisy spaces. Personalization also emerged as a gap; participants imagined glasses that adapt to their communication habits, sign preferences, or commonly used phrases over time. Suggestions included better face tracking, directional microphones, or context-sensitive translation to mitigate error risk and improve usability in dynamic settings.

Synthesis

This multi-phase evaluation, from the initial pilot to real-time roleplay to video-based interviews, illuminated both the promise and practical limitations of AR smart glasses for Deaf-hearing communication.

The pilot study with a Deaf professor provided critical accessibility and cultural feedback at an early stage. Her observations shaped the first round of system refinements, including increasing text contrast, improving gesture recognition for left-handed users, and removing culturally insensitive visual cues. Her suggestion to include visible indicators for system state (e.g., when the glasses are listening or speaking) laid the groundwork for many design discussions in subsequent phases.

Building on these improvements, Study 1 explored real-time interactions through a restaurant role play between hearing participants and a researcher simulating a Deaf customer. Quantitative survey results showed that while texting was rated higher in ease and comfort, the AR glasses offered greater engagement and felt more "conversational" to some users. Participants appreciated that the glasses allowed face-to-face interaction and supported gesture-based control. However, some still felt the system lacked clarity in transitions and timing. These insights confirmed that the glasses were promising for social settings but required refinement in real-time responsiveness and transparency.

Finally, Study 2 expanded the evaluation to a broader pool of hearing users through video-based interviews. Participants observed the system in use from both third-person and first-person perspectives and then reflected on usability, timing, visual feedback, and interaction flow. This study reaffirmed many of the pain points identified earlier, particularly the need for clear visual indicators, a more human-like voice output, and better timing customization. The inclusion of pre-interview expectations and post-video reflections allowed us to trace how mental models evolved after exposure, revealing a shift toward greater openness and confidence in engaging with Deaf individuals using AR technology.

Together, these studies form an iterative feedback loop that has already influenced the glasses' ongoing development. Each phase offered a different lens, accessibility in the pilot,

usability in Study 1, and broader expectation mapping in Study 2. Across all phases, users consistently emphasized the value of:

1. Clear visual indicators to support turn-taking
2. Natural, human-like voice synthesis
3. Flexible timing that adapts to context
4. Hybrid modes allowing both structured and freeform interaction
5. Cultural and emotional nuance in system feedback

As the glasses continue to evolve, these findings provide a roadmap grounded in real user experiences. Importantly, the studies highlight that inclusive communication technology must go beyond functional translation, it must also foster authenticity, comfort, and social reciprocity. By involving both Deaf and hearing perspectives across multiple contexts, our approach ensures that future design iterations are not only technically sound, but also socially resonant.

CHALLENGES

Several challenges emerged during both the design and evaluation phases of our project. Since the prototype was not fully developed, iterations faced both unique and shared challenges.

Pilot Study

1. Lack of certified interpreter present made the experiment and evaluation longer than anticipated
2. Limited access to diverse D/HoH individuals
3. Relying on Wifi made the system susceptible to unexpected interruptions, occasionally causing us to restart the simulation

The pilot study provided insightful design updates that we implemented to prioritize D/HoH perspectives. We changed a minor technical detail making signed inputs optional only if an initial option set is insufficient, as well as implemented UI/UX improvements that our participant suggested.

Study 1

1. Inconsistent gesture detection which caused delay and user confusion
2. Relying on Wifi made the system susceptible to unexpected interruptions, occasionally causing us to restart the simulation
3. The lack of a functioning real-time sign recognition module meant we had to manually

input signed responses, which may have influenced pacing and perceived authenticity.

These limitations impacted the realism of Study 1 and required us to adapt from early experiments by offering clearer pre-study demos and manual resets during breakdowns.

Study 2

1. Time constraints limited the length of demo videos presented to participants
2. A video demo may not be as digestible as a live demo

Shared Challenges

1. Inconsistent gesture detection may cause delay and user confusion
2. Limited scope of scenarios explored may have produced limited useful insights of both successes, failures, and future implications

These technical issues complicated our ability to assess true system usability under natural conditions, especially in terms of response speed and interaction fluidity. We addressed this by collecting both quantitative Likert data and qualitative feedback, which allowed us to triangulate participant perceptions despite system inconsistencies. Our findings will be used to inform iterative changes and more robust evaluation in future testing rounds.

EVALUATION OF SYSTEM

Our research employed an iterative, mixed-methods approach that combined a pilot with a Deaf professor, a live role-play experiment (Study 1), and a video-based interview study (Study 2). Together, these phases allowed us to triangulate usability, social, and attitudinal data while steadily refining the protocol. Across the three sessions we collected (1) in-situ behavioural observations, (2) post-task Likert ratings on clarity, comfort, social connection, and patience, and (3) semi-structured interview reflections. This layering of quantitative and qualitative evidence provided convergent validity: survey trends (e.g., higher ‘social engagement’ scores for the glasses condition) mapped onto open-ended comments that participants felt the interaction was ‘more human’ than texting. The multi-phase design also enhanced ecological validity: feedback from the expert pilot directly shaped later tasks, reducing the risk of culturally inappropriate cues and ensuring the subsequent studies probed realistic restaurant scenarios. Finally, repeating the restaurant script with both the glasses and a texting baseline within-subjects gave us a natural control, enabling more

confident attribution of participant preferences to the interaction modality rather than to scenario content.

Our system evaluation demonstrated that broad design research is crucial for the initial steps of a complex project, while recognizing that underdeveloped technology itself may cause gaps in research. By beginning with contextual inquiry and pilot role-plays, we surfaced the cultural nuances and situational constraints that shape Deaf-hearing interactions and distilled them into an ethically grounded study protocol. Yet once the prototype entered the lab, we noticed a blurred line between true usability barriers and mere engineering debt. As a result, some “frustration” ratings likely reflected hardware hiccups rather than fundamental flaws in the mediated conversation model, limiting the confidence with which we can generalise socio-emotional findings. The experience underscores a reciprocal rhythm: expansive design research sets the direction, but promptly maturing the critical technical components is essential.

FUTURE WORK

Our user studies revealed that many “usability” frustrations were actually engineering hiccups, such as manual sign entry, Wi-Fi delays, and brittle gesture detection obscured the true interaction model. The next research iteration should therefore prioritize a fully automated pipeline: on-device ASL recognition, low-latency speech-to-text, and real-time LLM response generation. Achieving this will let evaluators distinguish genuine socio-emotional barriers from mere system lag.

Participants wanted different pacing at a restaurant than they would in a personal conversation, and Deaf advisors stressed the need for culturally appropriate cues. Future work should explore adaptive timing and voice-tone models selected automatically from context signals such as ambient noise, GPS location, and conversation history.

Across studies, both Deaf and hearing users asked for unobtrusive signals showing when the glasses are listening, processing, or speaking. Future prototypes should prototype and A/B-test micro-LED patterns or haptic taps that satisfy three sometimes competing goals: (a) low visual clutter for the wearer, (b) legibility to the hearing partner, and (c) cultural appropriateness (avoiding gestures that clash with Deaf etiquette). Evaluating which indicators best support turn-taking in noisy, multi-party settings (e.g., airports) will push the work from controlled scripts toward ecological validity.

Current findings are based on four live role-plays and twelve video interviews, all with hearing participants except the pilot Deaf professor. To gauge sustained social impact,

future studies should (1) recruit a larger, diverse Deaf cohort, (2) run weeks-long diary or experience-sampling deployments in multiple venues, and (3) track longitudinal metrics such as trust, fatigue, and perceived autonomy. Such field trials will surface latent issues—wifi connection, social stigma, privacy optics—that short lab sessions cannot capture.

While the current focus is ASL-English, the underlying architecture could generalize to other signed-spoken pairs. Exploring transfer-learning strategies for low-resource sign languages would extend impact and force the design to accommodate different grammar orders and spatial referencing conventions.

Real-world deployment will intersect with ADA compliance, data-protection laws, and emerging standards for AI accessibility tools. A parallel research track should map regulatory requirements, prototype transparent data-consent flows, and engage Deaf-led organisations as co-designers to ensure the technology advances *with* rather than *for* the community.

We hope our mentor uses the findings explored in this paper to further develop this accessible communication.

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